LAMMPS official dem06-spin-identical

Spin preservation with friction inactive | 29 October 2026

PASS	Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag unvetted.

Official quantities, errors and acceptance limits

Quantity	Reference / limit	CPU dt	GPU dt	CPU dt/2	GPU dt/2
Preserved omega_y (rad/s)	22.86 relative limit 1e-08	22.86 err 0 PASS	22.86 err 0 PASS	22.86 err 0 PASS	22.86 err 0 PASS
Spurious vx (m/s)	0 absolute limit 1e-08	0 err 0 PASS	0 err 0 PASS	0 err 0 PASS	0 err 0 PASS
Spurious vy (m/s)	0 absolute limit 1e-08	0 err 0 PASS	0 err 0 PASS	0 err 0 PASS	0 err 0 PASS

Check time=0.00024 s; original step 2400. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

Original bodies and source-to-native mapping

Body	Radius / density	Initial xyz (m)	Initial v (m/s)	Initial spin (rad/s)
lower	R=0.0025 rho=2700	(0, 0, -0.002525)	(0, 0, 0.2)	(0, 22.86, 0)
upper	R=0.0025 rho=2700	(0, 0, 0.002525)	(0, 0, -0.2)	(0, -22.86, 0)

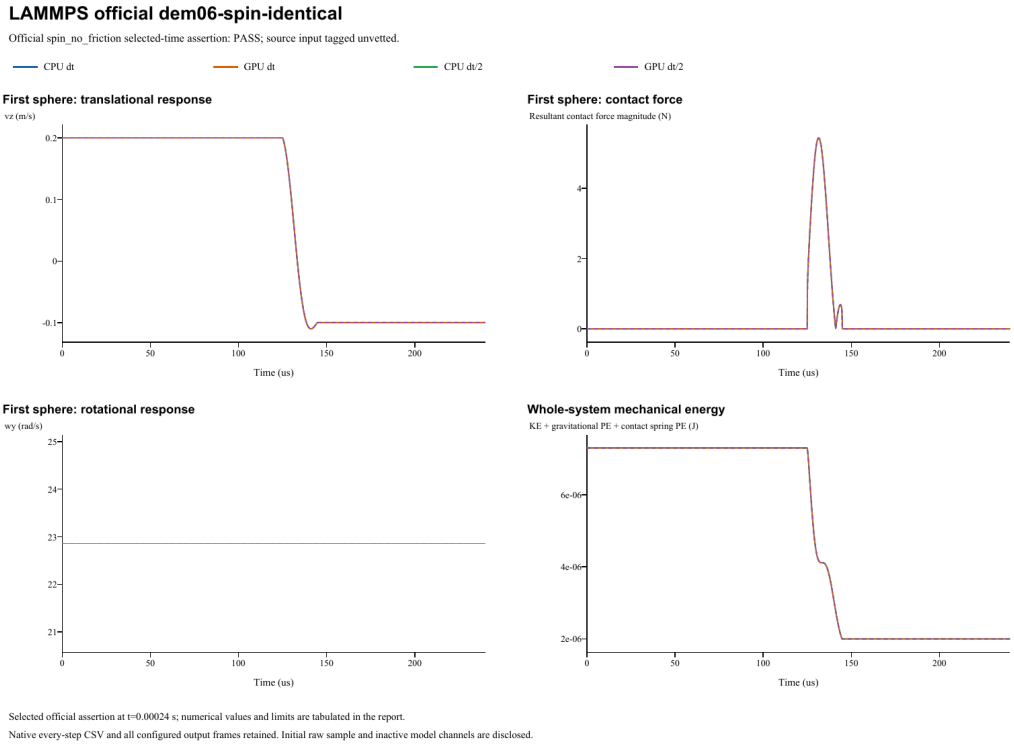
Hertz Fn_elastic=(4/3)*E_eff*sqrt(R_eff)*delta^1.5; native E_eff=3.927729772e+10 Pa. Stored linear kn is inactive. Automatic damping uses the harmonic material restitution and the active normal/tangential stiffness; stored manual dashpots are inactive. Tangential inputs: kt=0 N/m (explicit in linear mode), mu=0.4. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0.

Gravity=(0, 0, 0) m/s^2. Base/half dt=1e-07/5e-08 s; 2400/4800 steps. Equal opposite spins make relative tangential contact velocity exactly zero. The configured nonzero friction is retained; this is a zero-tangential-channel regression, not validation of active tangential dashpot equivalence.

Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob 97f00c537ce8d677399d1fc390e6e5f70605c32f. Official tag: unvetted.
LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

LAMMPS native histories and refinement



CPU/GPU and half-step comparison

Quantity	CPU dt error	CPU dt/2 error	Error type / limit
Preserved omega_y (rad/s)	0	0	relative: 1e-08
Spurious vx (m/s)	0	0	absolute: 1e-08
Spurious vy (m/s)	0	0	absolute: 1e-08

CPU/GPU whole-history maximum position / velocity / spin / force differences: 0 m / 4.34e-19 m/s / 0 rad/s / 8.88e-16 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 4.07e-20 kg m/s / 2.17e-19 m / 2.52e-17 m/s. Gravity is added separately to saved native contact forces.

Complete results and verified scope

All 14,404 native frames are retained; main frame counts=[2401, 2401, 4801, 4801]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.